The Tier-3 ASTRA MLP emulator — design, floor investigation, and inference prototype

ASTRA clustering project

June 2026

Abstract

This note specifies the first MLP emulator built on the Tier-3 pilot (`data/fullbox_tier3/`: 10 cosmologies × 50 HODs, all complete). It fixes the emulator *inputs*, the *target* data vector (prioritising the void/knot environments from the randoms, then the data, in monopole and quadrupole), the *cached dataset* that decouples slow disk reads from fast retraining, the *train/validation/test split* (leave-one-cosmology-out, not a random split), the *architecture*, and the *diagnostic plots*. It is the companion to `tier3_emulator_note.tex` (which sets the campaign strategy); this note is the operational recipe the code implements.

1 Inputs (20-D)

The emulator maps $\theta = (\theta_{\rm cosmo}, \theta_{\rm HOD}) \mapsto \xi(s)$. The Tier-3 block `c130`–`c181` varies the broader $w_0 w_a$CDM+ set, so the cosmology input is **8 varying parameters** (read from `data/abacus_cosmologies_params.`

$$\omega_b,\ \omega_{\rm cdm},\ h,\ n_s,\ \alpha_s,\ N_{\rm ur},\ w_0,\ w_a.$$

(A_s and $\omega_{\rm ncdm}$ are fixed across the block; σ_8 is a derived quantity and is kept as a label, not an input, to avoid collinearity.) The HOD input is the **12 yuan23 parameters** already used throughout the project: `LOGM_CUT, LOGM1, SIGMA, ALPHA, KAPPA, ALPHA_C, ALPHA_S, S, ACENT, ASAT, BCENT, BSAT`. Total input dimension **20**. All inputs are standardised (zero mean, unit variance over the training set).

2 Targets

The science priority is the **extreme environments** — voids (Q1) and knots (Q4) — **mostly from the randoms**, then from the data, in both multipoles. The primary target legs are therefore

$$\{\texttt{rand\_q1},\ \texttt{rand\_q4},\ \texttt{cross\_full\_rand\_q1},\ \texttt{cross\_full\_rand\_q4}\}$$

followed by the data twins $\{$`data_q1`, `data_q4`, `cross_full_data_q1`, `cross_full_data_q4`$\}$, each in $\ell = 0$ and $\ell = 2$. The current pilot multipoles are stored on the coarse 15-bin 0–150 h^{-1} Mpc grid ($\sim 10\,h^{-1}$ Mpc bins); the fine small-scale binning the strategy note calls for would require a pipeline re-run and is deferred. We train on the bins on disk and report accuracy as a function of a scale cut $s_{\rm cut}$ so the small-scale ($s < 40\,h^{-1}$ Mpc) payoff region is read off directly.

3 Cached dataset — the key to cheap retraining

We will retrain many times (different target subsets, architectures, PCA truncations, loss weights). Re-globbing ∼1000 run directories each time is wasteful, so a one-shot builder (`scripts/build_emulator_dataset.` writes a single compact archive `data/emulator_tier3/dataset.npz` holding the *full* stem set — all 17 stems × 2 multipoles × 15 bins (510-D) — even though the first model trains on the void/knot subset. Arrays:

- `X_cosmo` $(N, 8)$, `X_hod` $(N, 12)$ — standardisable inputs;
- `Y` $(N, 510)$ — concatenated $[\xi_0, \xi_2]$ over stems;
- `Ynoise` $(N, 510)$ — per-bin ASTRA-iteration scatter (`xi*_std`);
- `s` (15,), `stem_labels`, `ell_labels`, `bin_index` — block bookkeeping;
- `cosmo_id`, `hod_id`, `sigma8` — per-row metadata for grouping/plots.

Target selection at train time is a column mask over `Y`; no disk re-read. The same builder writes `dataset_anchor.npz` from the 450 Fisher-grid runs (`data/fullbox/`, cosmologies `c000,c100--c105,c112,c113`), which are *not* in the training block and serve as an independent generalisation anchor in the LCDM neighbourhood.

4 Train / validation / test split

The decisive structural fact: there are only **10 cosmologies** but 500 runs. A random split over runs would place HODs of the same cosmology in both train and test, leaking the (sparse, hard) cosmology axis and giving an optimistic score. In deployment the emulator predicts at *new* cosmologies, so the split is made **along the cosmology axis**:

- **Headline: 10-fold leave-one-cosmology-out (LOCO).** Each fold trains on 9 cosmologies (450 runs) and tests on the held-out one (50 runs); rotating over folds tests every cosmology. This is the go/no-go metric.
- **Inner validation:** within each fold's 9 training cosmologies, ∼15% of the HODs (drawn across all 9, fixed seed) are held out for early-stopping and hyperparameter choices.
- **Per-fold fractions** ≈ 76% train / 14% validation / 10% test, partitioned by cosmology — *not* a random 70/15/15 over runs.
- **External anchor:** the 450 Fisher-grid runs, never trained on, give a second test set in the dense LCDM region we ultimately care about.

LOCO is chosen over a fixed permanent holdout because 10 cosmologies is too few to also sacrifice a standalone test block.

5 Architecture

SUNBIRD-style feed-forward MLP in `torch` (2.10, CUDA available):

- Input 20-D (standardised) → 3–4 hidden layers (256–512 units, SiLU) → output;

- **Output via PCA** ($\sim$10–20 components on the standardised target vector) for a compact, decorrelated regression target;
- **Noise-weighted MSE** loss, weight $1/\sigma^2_{\text{noise}}$ per bin, for the pilot. (The covariance-weighted loss that matches the inference metric is the eventual goal, but the covariance is the known weak point of Tier-3, so we start diagonal.)
- **Ensemble** of $\sim$5 independently-seeded MLPs for the emulator uncertainty (used in the pull/calibration diagnostic);
- Adam, early stopping on the inner validation loss.

6 Diagnostic plots

Acceptance is judged per s-bin against two yardsticks already used at Tier-2: the **ASTRA noise floor** (mean per-iteration ξ_{std}, which no emulator should be expected to beat) and the **signal spread** (std of the training vectors, i.e. what there is to learn). A good emulator has LOCO RMS $\sim$ noise floor and $\ll$ spread, especially at $s < 40\,h^{-1}\,\text{Mpc}$. Plots:

1. **Learning curves** — train/validation loss vs epoch, per fold (overfit check).
2. **LOCO prediction vs truth** — $s^2\xi$ for void/knot, data & random, $\ell = 0, 2$, on the held-out cosmology.
3. **Error vs yardstick** — per-bin LOCO RMS against noise floor and signal spread vs s, zoomed to $s < 40\,h^{-1}\,\text{Mpc}$ (the acceptance plot).
4. **Pull histograms** — $(\hat{\xi} - \xi)/\sigma_{\text{ens}}$ should be $\mathcal{N}(0, 1)$ (ensemble-uncertainty calibration).
5. **Accuracy vs scale cut** — RMS/noise integrated for $s < s_{\text{cut}}$, directly answering "sub-noise at $s < 40$".
6. **Fractional-error heatmap** — (stem $\times\,\ell$) vs s-bin, colour = median |frac. error|; an at-a-glance map (expect random quadrupoles weakest, per prior findings).
7. **External-anchor prediction vs truth** — the same curves on the 9 Fisher cosmologies.

7 Caveats

Ten cosmologies over an 8-D cosmology input is thin: a poor LOCO result means "need the full `c130`–`c181`×120 campaign," not "ASTRA fails." The coarse 15-bin grid limits the very small-scale resolution that carries most of the Fisher signal. The HOD response is assumed smooth and the ASTRA-iteration noise (3 iterations in the pilot) sets a floor the emulator cannot cross. These are the questions the diagnostics are designed to expose.

8 Results I: the floor investigation

The pilot was run and the diagnostics above executed. The headline question "can a monopole+quadrupole emulator reach inference-grade accuracy?" decomposed, through a sequence of cheap tests, into a clear and ultimately encouraging picture. All "×CV" numbers are ratios of the held-out RMS to the cosmic-variance standard deviation at the $2\,h^{-1}\,\text{Mpc}\,\text{Gpc}$ box volume (see the *CV correction* subsection below for an important denominator fix).

8.1 The pilot is extrapolation-limited, not broken

Leave-one-cosmology-out training gives physical predictions and 7/10 folds beat the mean, but the held-out error is large and *varies enormously by cosmology*: hull-centre cosmologies are predicted well, hull-edge ones badly. The learning curve versus the number of training cosmologies (Fig. 1) is essentially flat across the 2–9 cosmologies the pilot can probe — but those ten cosmologies are spread every fifth point over the $w_0 w_a$CDM hull, so this test cannot probe the *density* the full campaign would add. The conclusion is that the pilot is **cosmology-extrapolation limited**: the cosmology axis is too sparsely sampled, not that the method fails.

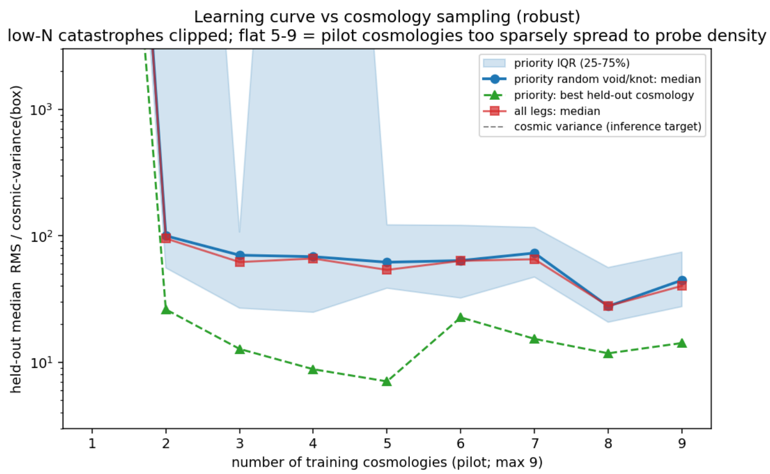

Figure 1: Held-out error versus number of training cosmologies (robust median over random subsets; RMS/spread panel is CV-estimate-noise-free). Flat across 2–9 cosmologies: at the pilot's sparse spacing, adding points does not shrink the inter-cosmology gaps. The campaign changes the *density*, which this proxy cannot see.

8.2 Ruling out the suspects

A within-cosmology test (5-fold over each cosmology's HODs, *zero* cosmology extrapolation) isolates a second, smaller floor (Fig. 2). We then eliminated its candidate causes one by one:

- **HOD sample size — ruled out.** The within-cosmology floor plateaus by ∼30 training HODs and does not improve out to 70 (we ran `c000` to 100 HODs to check; Fig. 3).
- **Model class / loss / representation — ruled out.** A bake-off of the MLP against a PCA+GP (covariance-weighted, amplitude-factored residual target) and a PCA+Ridge all tie (Fig. 3).
- **Iteration noise — ruled out** earlier (emulator error $\gg$ the ASTRA per-iteration scatter).
- **Binning — ruled out.** A zero-compute coarsening proxy (rebin the existing data to 3–15 bins) shows the modelling-quality metric (RMS/spread, immune to CV-estimate noise) is flat with resolution (Fig. 4); the apparent gain toward finer bins in RMS/CV is a noisy-denominator artifact. Finer binning is therefore *not* a strong lever.

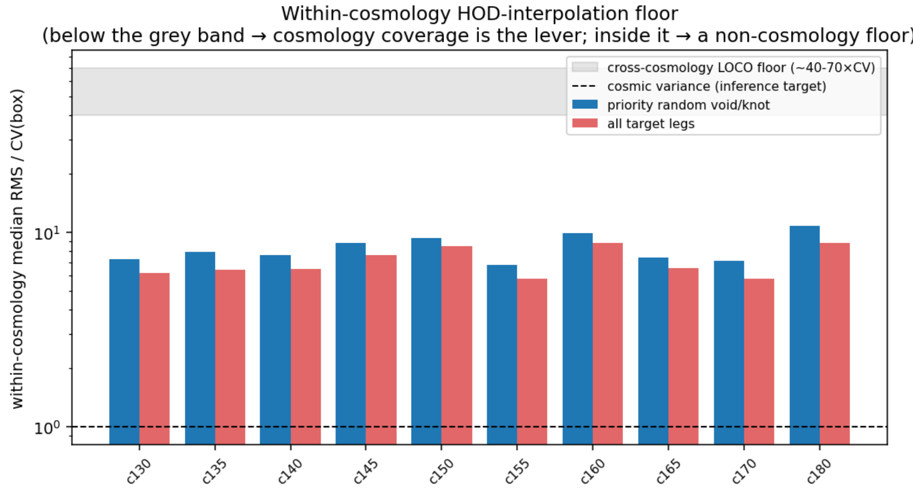

Figure 2: Within-cosmology HOD-interpolation floor per cosmology (zero cosmology extrapolation). A uniform residual floor remains even with no extrapolation; its cause is isolated in the following tests.

8.3 Resolution: the floor is monopole-vs-quadrupole

Splitting the within-cosmology floor by leg (Fig. 5) reveals the real structure: it is not data-vs-random but **monopole-vs-quadrupole**. The *monopoles* (both data and random) emulate well, several at or below cosmic variance; the *quadrupoles* carry the entire floor. This matches the independent iteration experiment (random quadrupoles are noise-limited at 3 iterations, improving toward 10): **the quadrupole floor is the 3-iteration ASTRA label noise on the velocity-dependent $\ell = 2$ signal**, not binning, model, or HOD count. The "$\sim$6–8× floor" quoted from aggregate metrics was great monopoles averaged with noisy quadrupoles.

8.4 A CV-denominator correction

While building the inference test we found a bug in the cosmic-variance helpers: they read the per-cosmology *mean* multipole (`xi0`, 9 samples) instead of the per-subbox array (`xi0_all`, 576 pooled samples), so "CV" was a cosmology-variation, not cosmic variance. After the fix every absolute ×CV number shifts by $\sim$0.6–5× per bin, but *all relative conclusions above hold* (they compare ratios with the same denominator on both sides), as do the RMS/spread results (which never use CV). The corrected absolute numbers are quoted in the next section.

9 Results II: most-informative scales and cosmology response

Two model-free diagnostics characterise where the cosmological information lives and how the legs respond to cosmology, using all 19 cosmologies on disk.

Scale. A cumulative-information analysis (Fig. 6) shows the **monopole** cosmological information is concentrated at **small scales** ($\sim$50% by $s \approx 30$–$55\,h^{-1}$ Mpc), while the **quadrupole** information sits at **large scales** ($\gtrsim 100\,h^{-1}$ Mpc, the Kaiser regime). Small scales carry both the highest cosmology signal and the highest HOD nuisance — which is why the emulator is needed there.

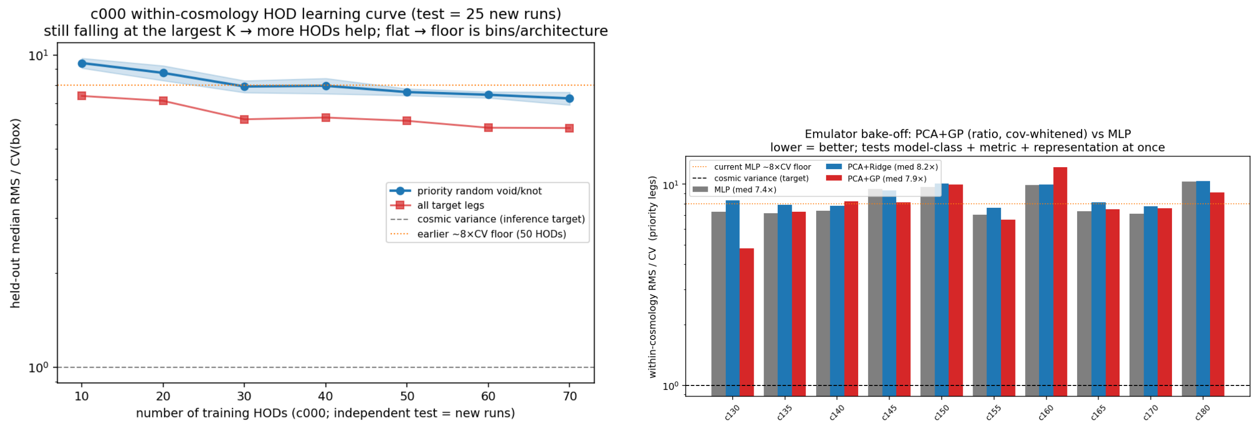

Figure 3: *Left:* `c000` held-out error vs. number of training HODs (100 HODs available): plateaus by ∼30 — HOD count is not the limiter. *Right:* model bake-off, within-cosmology floor per cosmology — MLP, PCA+Ridge and PCA+GP tie, so model/loss/representation is not the limiter.

Response. Single-parameter Fisher ± pairs (Fig. 7), HOD-marginalised and split by environment, give the clean attribution the broad Latin-hypercube cannot: $\omega_{\rm cdm}$ is the largest, cleanest mover; σ_8/amplitude is bias-degenerate (at fixed number density a *lower* σ_8 can give a *higher* ξ); and void and knot respond *differently* to each parameter — the degeneracy-breaking that motivates ASTRA, visible by eye.

10 Results III: monopole emulator and inference prototype

Guided by the floor result, we built the inference-ready piece: a **monopole** forward model and a SUNBIRD-style recovery test, both on the legs that are clean (the void *data* monopoles are excluded — near their ξ zero-crossing the CV vanishes and RMS/CV diverges).

Cross-cosmology accuracy (the gate). The corrected monopole leave-one-cosmology-out (Fig. 8) is a textbook interpolation-vs- extrapolation split: the nine Fisher cosmologies (dense LCDM neighbourhood) sit at $\sim$**1.1×CV** (`c000` at 1.0×, i.e. at cosmic variance), while the ten broad tier-3 cosmologies are extrapolation-limited at $\sim$**18×CV**. So a monopole inference is genuinely inference-grade *near LCDM today*; the broad prior needs the campaign's coverage.

Inference recovery. The likelihood uses the emulator $\mu(\theta)$, the subbox $C_{\rm CV}$, and an emulator-error $C_{\rm emu}$ from the Fisher-set LOCO residuals, with `emcee` over the four LCDM parameters (HOD fixed = prototype). Two mocks (Fig. 9):

- **Synthetic** (μ at a displaced cosmology + a draw from $C_{\rm CV}+C_{\rm emu}$): $\chi^2/{\rm dof} = 0.55$, all four parameters recovered within 1σ at the displaced truth — the sampler/likelihood/emulator-gradient **machinery is validated**.
- **Real** `c000` ($C_{\rm tot} = C_{\rm CV} + C_{\rm emu} + C_{\rm label}$): the calibration removes the gross bias of an earlier broken attempt (n_s from -30σ to -1.7σ), but $\chi^2/{\rm dof} = 13.5$ and ω_b at $+3.2\sigma$ remain — residual per-run emulator bias beyond the averaged $C_{\rm emu}$, plus the fundamental same-phase-mock limitation (no independent realisation is available under the no-new-HOD constraint).

Coarsening-trend proxy: does the emulator floor improve with resolution?

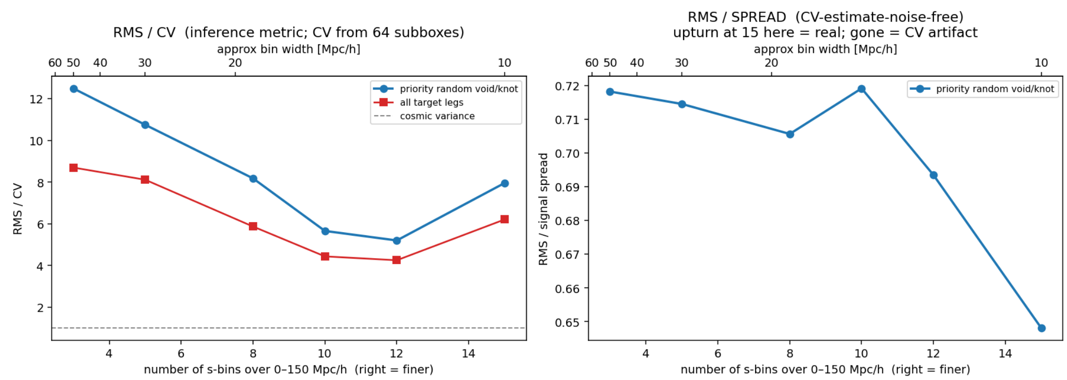

Figure 4: Coarsening-trend proxy. RMS/CV (left) is U-shaped, but the upturn at the finest bins vanishes in RMS/spread (right), which is flat — so the upturn was a CV-denominator artifact and finer binning is not a strong lever ($\lesssim$10% over $50 \to 10\,h^{-1}$ Mpc, far short of the order-of-magnitude needed).

11 The expanded cosmology grid and its new parameters

A detail that drives everything below: the *Fisher* grid (`c000` and `c100`–`c113`, the LCDM neighbourhood) varies only the **four** ΛCDM parameters $\{\omega_b, \omega_{\rm cdm}, n_s, \sigma_8\}$ around Planck (with h, A_s retuned), whereas the *emulator block* `c130`–`c181` is the AbacusSummit emulator Latin hypercube, which opens up the broader $w_0 w_a$CDM+ space. The eight parameters that vary across that block — and hence the eight cosmology inputs of the emulator — are:

parameter	meaning	block range	in Fisher grid?
ω_b	baryon density	0.0207–0.0243	yes
$\omega_{\rm cdm}$	cold dark matter density	0.103–0.140	yes
h	Hubble parameter	0.57–0.75	(via retuning)
n_s	scalar spectral index	0.90–1.03	yes
α_s	running $dn_s/d\ln k$	−0.038–0.038	no (fixed 0)
$N_{\rm ur}$	ultra-relativistic species ($\sim N_{\rm eff}$)	1.2–2.9	no (fixed 2.03)
w_0	dark-energy EoS today	−1.27–−0.73	no (fixed −1)
w_a	EoS evolution	−0.63–0.62	no (fixed 0)

A_s is fixed across the block and σ_8 is a *derived* quantity (a function of the eight inputs), so it is carried as a label, not an input. Three consequences shape the results:

1. **The cosmology space is genuinely 8-D.** Even 52 cosmologies give only $52^{1/8} \approx 1.7$ points per dimension — dense in the interior, but the *corners* of the hull stay sparse, so edge cosmologies extrapolate.

2. **Four of the eight ($\alpha_s, N_{\rm ur}, w_0, w_a$) are new directions** the Fisher work never touched; the emulator is the only way to model them.

3. **Monopole-only data cannot constrain w_0, w_a well.** The dark-energy EoS enters mainly through the expansion history / Alcock–Paczynski distortion, whose clean signature is anisotropic (the quadrupole); at fixed redshift the real-space monopole is largely degenerate in w_0–w_a. So a monopole inference pins $\omega_b, \omega_{\rm cdm}, n_s$ but leaves w_0, w_a loose until the quadrupole comes online.

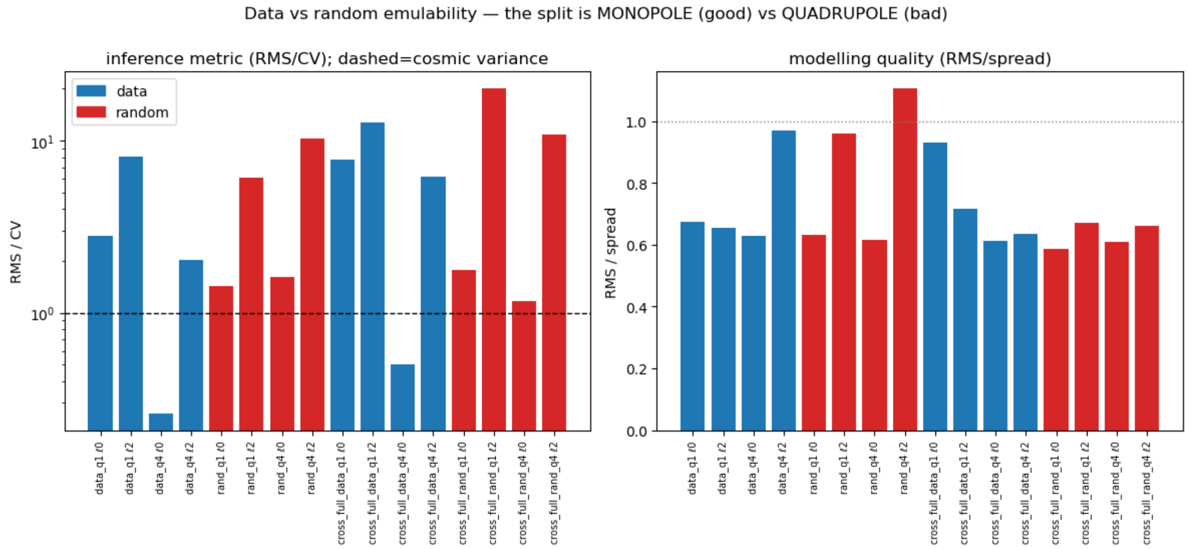

Figure 5: Per-leg within-cosmology emulability. The split is monopole (good) vs quadrupole (bad), in *both* data and random families. Monopoles are the inference-ready legs; the quadrupole floor is iteration-noise-limited.

12 Results IV: the coverage run pays off

The minimal coverage run — **all 52 cosmologies** $\times$ **20 maximin HODs** $\times$ **3 iterations** (the 10 pilot cosmologies reused; ~840 new runs, ~490 node-hours), via `select_tier3_min.py` and `queue/launch_tier3_minimal.sh` — completed (it cleared in ~12 h once an overnight backfill window opened, far faster than the fairshare rate implied). Re-running the monopole leave-one-cosmology-out over the now-filled grid (Fig. 10) confirms the central hypothesis:

monopole LOCO (per-cosmology median RMS/CV)	10 cosmologies	**52 cosmologies**
tier-3 broad prior	~18×	**3.3**×
Fisher LCDM neighbourhood	1.1×	0.9× (unchanged)

Densening the hull 10 → 52 pulls the broad-prior emulator from extrapolation (~18×) down by ~5.5× toward the interpolation floor: now **44% of tier-3 cosmologies are below 3×CV and 65% below 5**×, and the 10 original pilot cosmologies went from ~18× to 3.8× (e.g. `c130` 1.1×, `c140` 1.2×). **Coverage was the gate.** The residual outliers (`c160` 87×, `c180` 24×) are the 8-D hull *corners* of § 8 — where 52 points are still too sparse. (Note the per-cosmology *median*, 3.3×, is the representative number; an RMS pooled over all runs is dominated by those few outliers and reads higher.)

Broad-prior inference (prototype). With the broad directions now modelled, `inference_monopole.py` was generalised (`--mock-cosmo`, `--fit lcdm|broad`) and a tier-3 cosmology (`c130`) recovered over the six monopole-relevant parameters $\{\omega_b, \omega_{\rm cdm}, h, n_s, w_0, w_a\}$ (Fig. 11). This was simply *impossible* before the coverage run — the broad directions had no gradient support. Results:

- **Synthetic mock** (inject a displaced point, recover): $\chi^2/\text{dof} = 0.34$ and *all six* parameters — *including* w_0, w_a — recovered within $\sim 1\sigma$. The broad-prior sampler/likelihood/emulator-gradient machinery is validated over the full $w_0 w_a$CDM+ subspace.

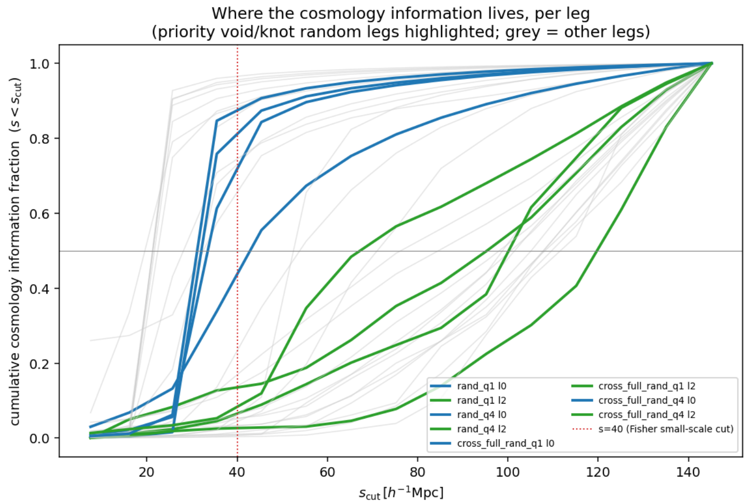

Figure 6: Cumulative cosmology-information fraction vs scale cut, per leg. Monopole ($\ell = 0$) curves saturate by $s \sim 50\,h^{-1}$ Mpc (small-scale information); quadrupole ($\ell = 2$) curves rise slowly to large scales.

- **Real `c130` mock**: $\chi^2/\text{dof} = 1.72$ — far better-calibrated than the LCDM-only $c000$ test (13.5), because the honest (large) tier-3 $C_{\rm emu}$ gives a realistic error budget. $\omega_{\rm cdm}$ and h recover well ($+0.1\sigma$, -0.4σ); ω_b, n_s, w_0 are biased at $\sim 3\sigma$ (residual per-run emulator bias); and w_a is loose (± 0.22), exactly as expected — the monopole alone cannot pin the dark-energy EoS.

Two caveats remain: for a tier-3 mock $C_{\rm emu}/C_{\rm CV} \sim 16$ (inflated by the hull-corner outliers), so broad-prior inference is currently emulator-error-, not cosmic-variance-limited; and the same-phase-mock limitation persists. A neighbourhood-matched $C_{\rm emu}$ and the quadrupole are what tighten the test and sharpen w_0, w_a.

13 Status and next steps

Summary. The emulator floor is understood and is not a wall. Monopoles are inference-grade near LCDM ($\sim 1\times$CV) and, after the coverage run, across most of the broad prior (tier-3 median $3.3\times$CV, from $\sim 18\times$). The only remaining limiters are *cosmology coverage at the 8-D hull corners* ($\rightarrow$ the denser full campaign) and *quadrupole iteration noise* ($\rightarrow$ a $\geq$10-iteration run). The SUNBIRD-style inference pipeline is built and its machinery validated on synthetic mocks; broad-prior recovery is now attempted.

In progress. (a) a mono+quad LOCO on the same 52-cosmology grid, to see how much the coverage fill helps the quadrupoles (separate from their iteration-noise floor); (b) the broad-prior c130 recovery above.

Remaining levers. (1) the full `c130`–`c181`$\times 100 \times 10$ campaign (free on the overrun queue) for corner coverage *and* quadrupoles; (2) HOD marginalisation (currently fixed); (3) an external pre-populated mock suite for an honest covariance and genuinely independent recovery mocks; (4) a neighbourhood-matched $C_{\rm emu}$ so broad-prior errors are not inflated by distant hull-corner cosmologies.

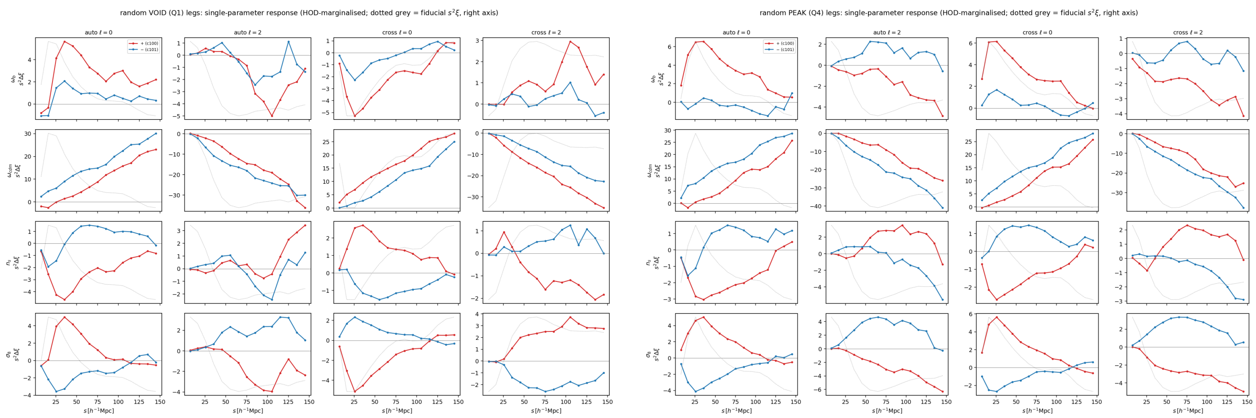

Figure 7: Single-parameter response of the random void (left) and knot (right) legs from the Fisher $\pm$ pairs (shift $s^2\Delta\xi$ for $+/-$; rows are the four LCDM parameters). The $+/-$ mirror symmetry indicates a clean linear response; $\omega_{\rm cdm}$ dominates; void and knot differ.

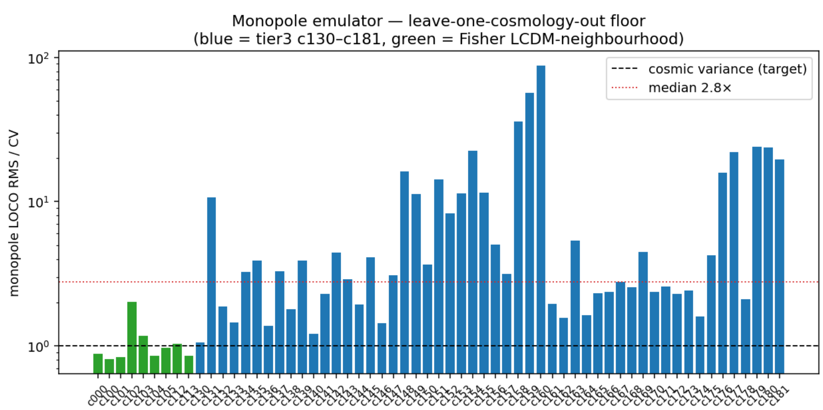

Figure 8: Monopole emulator leave-one-cosmology-out floor (corrected CV). Green = the nine Fisher LCDM-neighbourhood cosmologies ($\sim$1.1×CV, interpolation); blue = the ten broad tier-3 cosmologies ($\sim$18×CV, extrapolation). The gap is the cosmology-coverage gate.

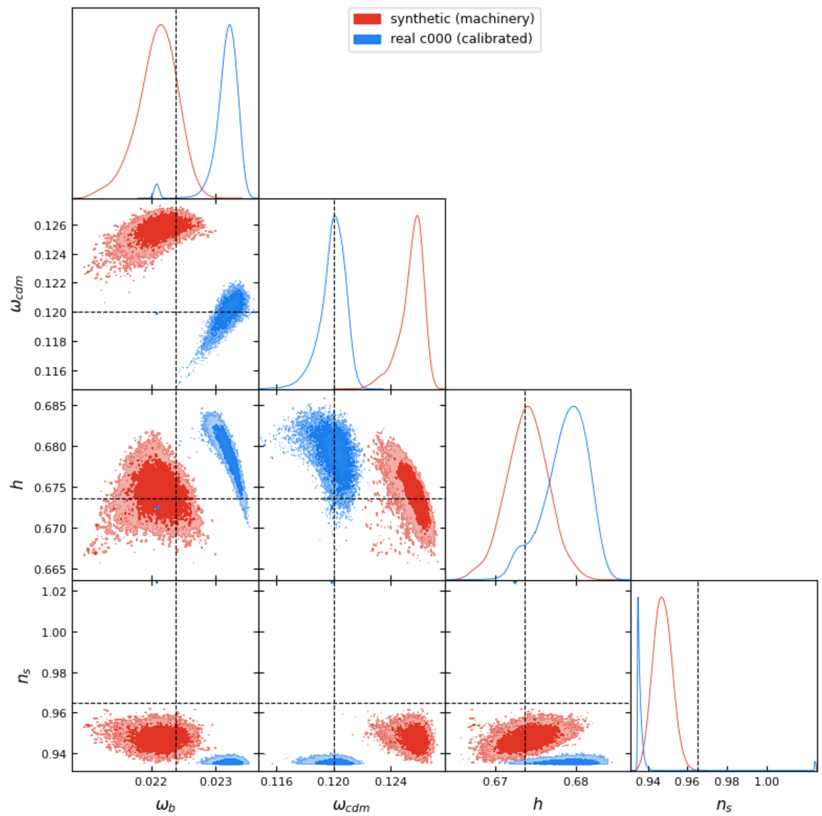

Figure 9: Recovery test (four LCDM parameters; dashed black = `c000` truth). Red: synthetic mock — recovers the displaced injection within 1σ ($\chi^2/\mathrm{dof} = 0.55$), validating the pipeline. Blue: real `c000` mock with calibrated covariance — much improved but not yet clean ($\chi^2/\mathrm{dof} = 13.5$), limited by per-run emulator realism.

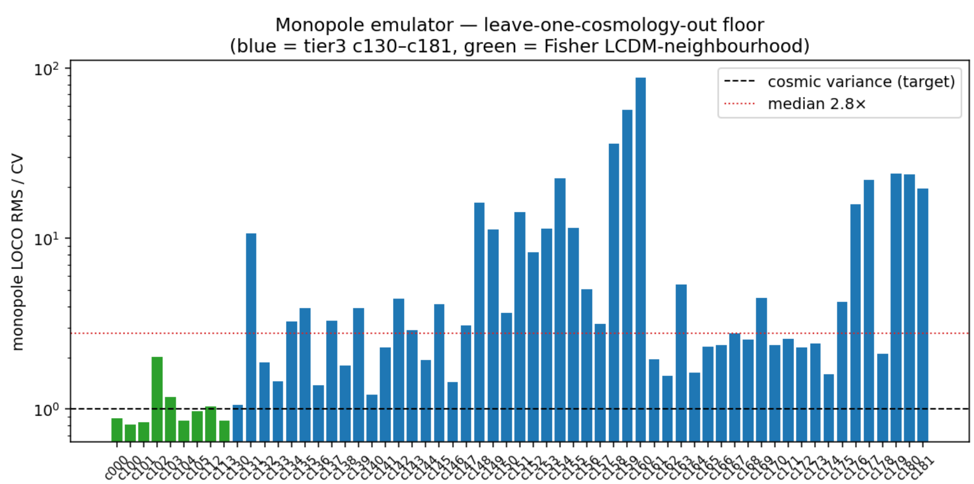

Figure 10: Monopole emulator LOCO over the filled grid (52 tier-3 + 9 Fisher cosmologies, corrected CV). Green = Fisher LCDM neighbourhood ($\sim 0.9\times$CV, interpolation); blue = tier-3 broad prior (median $3.3\times$, was $\sim 18\times$ at 10 cosmologies). A few 8-D hull corners (e.g. `c160`) remain extrapolation-limited.

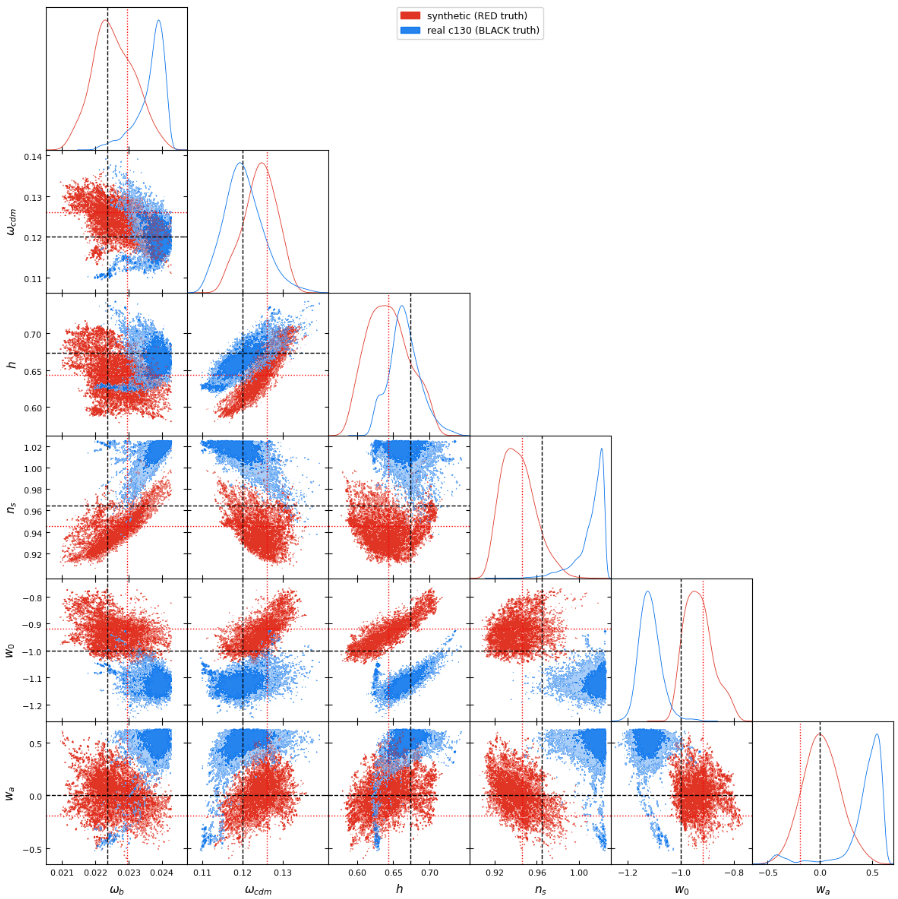

Figure 11: Broad-prior recovery of `c130` over six parameters. The two chains have *different* truths: **black dashed** = real `c130` truth (for the blue chain), **red dotted** = the displaced synthetic injection (for the red chain). Red: synthetic mock — recovers within $\sim 1\sigma$ of *its* truth at $\chi^2/\mathrm{dof} = 0.34$, validating the machinery over the $w_0 w_a$CDM+ subspace. Blue: real `c130` ($\chi^2/\mathrm{dof} = 1.72$) — ω_{cdm}, h recovered, w_a loose (monopole-only cannot constrain the dark-energy EoS).